ESTEFANY ARGUETA HERRERA

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EDUCATION

University of Massachusetts Amherst — Amherst, MA
Ph.D. Candidate in Organismal and Evolutionary Biology — Feb. 2021 - Present
Advisor: Lisa Komoroske

Iowa State University — Ames, Iowa
B.S. in Animal Ecology Emphasis: Fisheries and Aquatic Sciences — May 2020

PUBLICATIONS

Lonthair JK, NC Wegner, BS Cheng, NA Fangue, MJ O'Donnell, AM Regish, JD Swenson, **E Argueta**, SD McCormick, BH Letcher, and LM Komoroske. 2024. The gill oxygen limitation hypothesis remains unsupported by experimental evidence. Response to 'Testing mechanistic theories must be based on correct interpretations', Journal of Experimental Biology, 227, jeb248173.

Lonthair JK, NC Wegner, BS Cheng, NA Fangue, MJ O'Donnell, AM Regish, JD Swenson, **E Argueta**, SD McCormick, BH Letcher, and LM Komoroske. 2024. Smaller body size under warming is not due to gill oxygen limitation. Journal of Experimental Biology, 227 (4): jeb246477.

GRANTS AND FELLOWSHIPS

Total Funding Awarded: **$225,310.50**.

- UMass Amherst, College of Natural Sciences Teaching Fellow — Spring 2026
- UMass Amherst, Armstrong Graduate Student Research Grant **$22,400** — Spring 2026
- International Sea Turtle Society Symposium Travel Grant — Spring 2026
- UMass Amherst, Graduate School Spring 2025 Fieldwork Grant **$4,724** — Spring 2025
- UMass Amherst, Graduate Student Government (GSG) Research and Travel Grant **$400** — Spring2025
- UMass Amherst, Office of Inclusion and Engagement Travel Grant **$400** — Spring2025
- UMass Amherst, College of Natural Sciences, Diversity Travel Grant **$2,000** — Fall 2024
- UMass Amherst, Richard Cronin Fisheries Research Award **$1,386.50** — Spring 2024
- NSF Graduate Research Fellowship **$144,000** — Spring 2022
- UMass Amherst, Spaulding-Smith Fellowship **$50,000** — Spring 2021

RESEARCH EXPERIENCE

Graduate Dissertation Research — 2021 – Present
Molecular Ecology and Conservation Lab, University of Massachusetts, Amherst MA
Focus: Thermal Tolerance and adaptation in Ectotherms
Specific Skills:

- Conduct kinship analysis and estimate of population sex ratios for a Brazilian Green Sea Turtle Population
- Investigate the range expansion of Black Sea Bass in response to climate change using metabolic rate and thermal tolerance approaches
- Understand biological hierarchical responses to warming temperatures within ectotherms

Graduate Research Assistant — Sep 2021 to Present
Focus: Smaller body size under warming is not due to gill oxygen limitation

- Performed fish husbandry for a long-term growth and metabolic rate experiment on brook trout (Salvelinus fontinalis).
- Assisted in and conducted respirometry trials (RMR, MMR, PCrit) for fish at various temperatures.

Focus: Mating systems as mechanisms for resilience of species with temperature-dependent sex determination under environmental change

- Performed nighttime sea turtle monitoring on a remote Brazilian island consecutively for three months for two years.
- Taught undergraduates DNA extraction techniques and other genetic protocols in a molecular lab.

Undergraduate Research Assistant

Focus: Painted Turtle (Chrysemys picta) Nest Depth Thermal Variation Sep 2018 to Aug 2019

- Analyzed the relationship between nest depth and temperature using R, revealing critical temperature variations across nest regions that may influence sex determination in developing turtles

Focus: The Relationship of Body and Clutch Size in Loggerhead Turtles, on NC beaches May to Aug 2019

- Conducted nightly sea turtle monitoring along NC beaches in collaboration with NC Division of Parks and Recreation, documenting nesting patterns and collecting biological data
- Analyzed 40-year longitudinal dataset using R to identify significant correlation between maternal body length and clutch size in Loggerhead turtles

Focus: Cultural Empathy within Mentor and Mentee Relationships between underrepresented Graduate Students and Faculty Aug 2018 to May 2020

- Created and analyzed surveys on cultural empathy between graduate students and faculty members to improve experiences of underrepresented students in graduate programs

Focus: A Comparison of Reproductive Development Between Alaskan and Oregonian Rockfish Populations Jul to Aug 2018

- Partnered with charter boats to collect and analyze rockfish otoliths, comparing age and growth patterns between Oregon and Alaska populations through ring counting analysis

TEACHING AND MENTORSHIP

Graduate Teaching Assistant, University of Massachusetts, Amherst, Fall 2024 – Present

Introductory Physiology, BIO 288 Graduate Teaching Assistant: 150 Students, led weekly review sessions, graded assignments and exams, supervised 10 undergraduate teaching assistants.

Genomics and Data Analysis, BIO 479D Graduate Teaching Assistant: 38 Students, facilitated R-based labs, reviewed lab coding assignments, graded assignments

Invited Scientist and Math Teaching Assistant, Twice as Smart, Greenfield MA Dec 2024- Present

Invited Scientist, 25 Students – one-room schoolhouse K-12 Ecology based learning lessons

Math Teaching Assistant: Develop problem sets and learning tools for students in K-12. Focused on 8-12th grade math concepts.

Research Mentorship

Michaela Kirby: Café Scholars Undergraduate Internship, UMass Amherst June 2024– May 2026

Skills Developed: Intermittent respirometry, fish husbandry, net seining, snorkel survey, R Statistical analyses, literature review, scientific poster creation, scientific poster presentation

Cammi Cooper: Undergraduate Internship, UMass Amherst June 2025– December 2025

Skills Developed: Thermal Limits Testing, Research Design, fish husbandry, R Statistical analyses, literature review, scientific poster creation, scientific poster presentation

Amanda Furbacher: Café Scholars Undergraduate Internship, UMass Amherst June 2025– February 2026

Skills Developed: Thermal Limits Testing, Research Design, fish husbandry, R Statistical analyses, literature review, scientific poster creation, scientific poster presentation

Ana Stucker: Café Scholars Undergraduate Internship, UMass Amherst July 2023-December 2023

Skills Developed: Sea Turtle DNA Extractions, R coding skills

Highschool Student:
Neishaly: Dial Self Youth and Community Services, Greenfield MA March 2024
Skills Developed: Basic Life skills, self-autonomy, methods for self-motivation, self-discipline and resources for alternative school programs and learning.

PRESENTATIONS

Gloucester Marine Station Opening at UMass, Gloucester MA, *How temperature changes the 'energy budget' of Black Sea Bass* (Poster) Jun 2026
Evolution Course at Smith College, Northampton MA, *Hot fish in Cold Waters: Unraveling the mechanisms of thermal adaptations in Range-shifting Black Sea Bass* (Oral) Apr 2026
International Sea Turtle Society Symposium, Kona HI, *Can mating behaviors buffer climate impacts? Evidence from Fernando de Noronha Green Sea Turtle reproductive dynamics* (Oral) Mar 2026
Ecophysiology Course at UMass Amherst, Amherst MA, *Hot fish in Cold Waters: Unraveling the mechanisms of thermal adaptations in Range-shifting Black Sea Bass* (Oral) Nov 2025
Underwater In Salem Sound Lecture Series, Marblehead MA, *Swimming North: How Fish Adapt to Climate Change* (Oral) May 2025
Central Campus, Des Moines IA, *From Genes to Populations: Assessing Climate Change Impacts on Ectotherms* (Oral) May 2025
Twice as Smart, Greenfield MA, *Nature's Web: Exploring the Amazing World of Ecology!* (Oral) Jan 2025
Youth Climate Action Network (YCAN), Amherst MA, *A need for Climate Change Youth Action in K-12 Learning* UMass EducArticle (Oral) Nov 2024
Texas National McNair Research Conference, Denton TX, *Painted Turtle* (*Chrysemys picta*) *Nest Depth Thermal Variation* (Oral) Feb 2020
Duke Marine Lab REU, Beaufort NC, *The Relationship of Body Size and Clutch Size in Loggerhead Turtles, Caretta caretta, on North Carolina Beaches* (Oral) Aug 2019
Annual McNair Symposium, Ames IA, *Painted Turtle* (*Chrysemys picta*) *Nest Depth Thermal Variation* (Poster) Apr 2019
Hatfield Marine Science Center, Newport OR, *Phylogeny of Gastropoda*, (Oral) Aug 2018

HONORS AND AWARDS

Graduate awards and Grants
- International Sea Turtle Society Archie Carr Best Biology Oral Presentation Spring 2026
- UMass Amherst, CNS Outstanding Achievement Award Spring 2022

Undergraduate awards and Grants
- Outstanding First Year McNair Scholar Spring 2019
- Chuck and Marguerite Tice Fisheries and Wildlife Scholarship Spring 2019
- Iowa Chapter American Fisheries Society Scholarship Spring 2019
- Ronald E. McNair Scholar 2018-2020
- Iowa State University Dean's List F2015, S2016, S2018, F2018, S2019, S2020
- Cornell University Preview Weekend Spring 2018
- Inaugural George Washington Carver Spirit of Innovation and Service Award Fall 2016

RESEARCH AND TECHNICAL SKILLS

Field Techniques: Genetic tissue sampling, aquatic respirometry trials, sea turtle nesting monitoring, fish identification, otolith/statolith removal and reading, VIE tagging, MS-222 anesthetic delivery, vertebrate

dissection, remote fieldwork, sea turtle egg collection, sea turtle nest inventory, flipped tagging, PIT Tag deployment and monitoring, beach seining, purse seine netting, line fishing, AAUS Scientific Diver
Genomic Techniques: DNA and RNA Extractions and Sequencing, transcriptomic analysis, bioinformatics approaches
Software: R Statistics Software, parametric and non-parametric analyses (including mixed models); proficient working in the bash environment for bioinformatic analysis using Unity High-Performance Computing Cluster

RESEARCH WORKING GROUP PARTICIPATION AND MEMBERSHIPS

Marine Turtle Hatchling Sex Determination Working Group	2022-Present
UMass Behavioral, Animal Biomechanics, Morphology, Physiology, Ecology and Evolution (BAMPHEE) Lab Consortium	2023-Present
Society for the Advancement of Chicanos and Native Americans in Science (SACNAS)	2018 to Present

WORKSHOPS

Emerging Genomic Tools for Sea Turtle Biology & Conservation, workshop attendee	Mar 2026
Workshop on Molecular Evolution, workshop attendee	May 2025

INTERNATIONAL EXPERIENCE

Graduate Research Assistant, UMass Amherst	Fernando de Noronha, Brazil
Fundacao Projeto Tamar – Sea Turtle Monitoring	Spring 2022, Spring 2023
Study Abroad, Universidad Católica de la Santísima Concepción	Concepción, Chile
Ecology, Biodiversity, & Conservation Biology in Chile	Fall 2019

OUTREACH AND SERVICE

U.S. Army Reserve Officer, Army Captain	US Army Reserves
Human Resources Officer in 403rd Civil Affairs Battalion	2024 to Present
Human Resources Officer in 9-95th QM Battalion	2020 to 2024
Emergency Housing and overnight teen warming center staff	Greenfield, MA
DIAL/SELF Youth and Community Services	2022 to Present
Science Bound Mentor	Iowa State University
STEM Grad School Panel Member	Feb 2024, Oct 2024
CyCLIMB mentor	Dec 2021, Jan 2024
Coach	Franklin County YMCA
Girls on the Run	Fall 2023
Chapter Secretary	UMass Amherst
Society for the Advancement of Chicanos and Native Americans in Science Chapter	2021 to 2022
Mentor	UMass SACNAS
STEM Highschool Outreach event 2022	Apr 2022
L.I.N.K.S Mentor	MCAS Cherry Point
Lifestyle Insights, Networking, Knowledge, and Skills mentor in the Marine Corps	2021